	Earth's surface.	
3(a)(i)	The internal energy of a system is the sum of a random distribution of kinetic and potential energies associated with the molecules of a system.	[1]

No.	Solution	Remarks
3(a)(ii)	$\Delta U = Q + W$ ΔU – increase in internal energy Q – heat supplied to the system W – work done on the system	[1] equation [1] correct definition of symbols
3(a)(iii)	There is work done by gas against the atmosphere during expansion for gas at constant pressure. Using $\Delta U = Q + W \rightarrow Q = \Delta U + W_{BY}$, amount of heat supplied has to be larger for the same amount of increase in internal energy. Since ΔT depends on ΔU, therefore a larger amount of heat must be supplied to have the same amount of ΔT.	[1] [1] [1]
3(b)	Let θ be the equilibrium temperature. $m_w\, c_w\, (90 - \theta) = m_{ice}\, L_f + m_{water\ at\ 0\ °C}\, c_w\, (\theta - 0)$ $(0.5)\,(4200)\,(90 - \theta) = (0.1)\,(3.34 \times 10^5) + (0.1)\,(4200)\,(\theta)$ $\theta = 61.7$ °C	[1] substitution [1] answer
4(a)	Simple harmonic motion is defined as an oscillatory motion in	